LESSON 01 / 10

The derivative is a limit of secant slopes

If the limit exists, the derivative of f at x is

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{z\to x}\frac{f(z)-f(x)}{z-x}.$$

The quotient is the slope of the secant through $(x,f(x))$ and $(x+h,f(x+h))$. Its limiting value is the tangent slope and the instantaneous rate of change. A derivative must be a **finite** limit for ordinary differentiability.

WORKED EXAMPLE

Expand, cancel h, then take the limit

For $f(x)=x^2$ and $h\neq 0$,

$$\frac{(x+h)^2-x^2}{h}=2x+h\longrightarrow 2x.$$

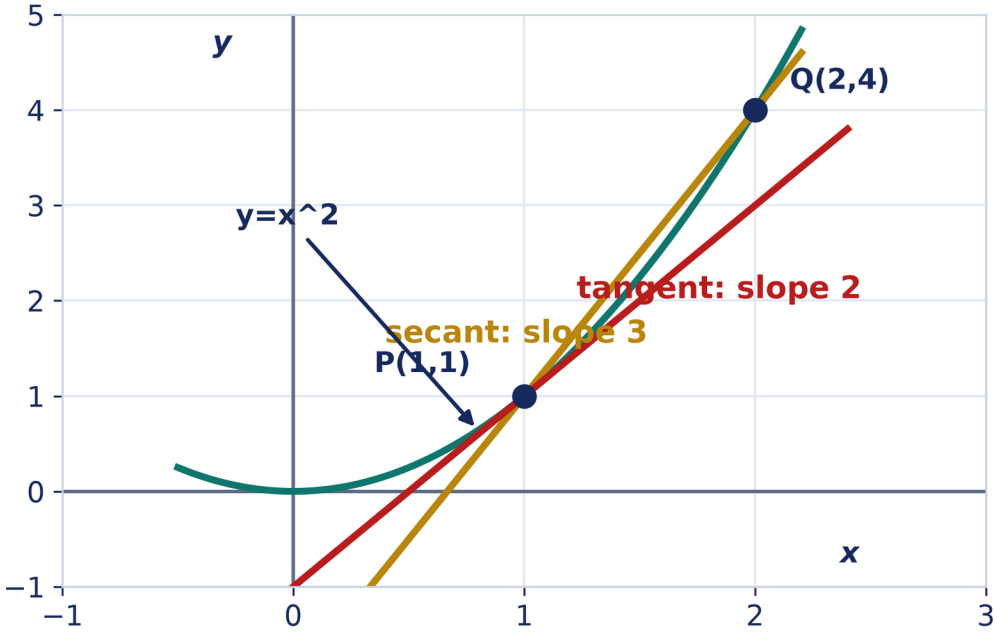
Similarly, for $x\neq 0$ and $x+h\neq 0$,

$$\frac{1/(x+h)-1/x}{h}=-\frac{1}{x(x+h)}\longrightarrow -\frac{1}{x^2}.$$

At $x=2$, the cubic quotient gives

$$\frac{(2+h)^3-8}{h}=12+6h+h^2\longrightarrow 12.$$

The secant slope approaches the tangent slope



READ THE FIGURE

For h=1 the secant slope is (4-1)/(2-1)=3. As Q approaches P, the secant slope tends to f'(1)=2 and the line tends to y=2x-1.

Differentiability forces continuity, but not conversely

If $f'(a)$ exists, then

$$f(x) - f(a) = \frac{f(x) - f(a)}{x - a}(x - a) \longrightarrow f'(a) \cdot 0 = 0.$$

Thus f is continuous at a. The converse is false: $|x|$ is continuous at 0 but its left and right difference quotients are -1 and 1.

The tangent uses a point and the derivative slope

At $(a, f(a))$, a nonvertical tangent is

$$y - f(a) = f'(a)(x - a).$$

For $y = x^2$ at $a = 1$, the point is $(1, 1)$ and the slope is 2, so the tangent is $y - 1 = 2(x - 1)$, or $y = 2x - 1$. Advanced tangent, normal and curve-sketching problems are developed in Chapter 19.

COMMON TRAP

Recognize a derivative before calculating a new limit

The limit

$$\lim_{h \to 0} \frac{\ln(e + h) - 1}{h}$$

is $f'(e)$ for $f(x) = \ln x$, hence it equals $1/e$. Matching $f(a + h) - f(a)$ is often faster than applying L'Hôpital's rule.

LESSON 02 / 10

Two one-sided slopes must both exist and agree

At a junction a, define

$$f'_{-}(a) = \lim_{h\to 0^{-}} \frac{f(a+h)-f(a)}{h}, \qquad f'_{+}(a) = \lim_{h\to 0^{+}} \frac{f(a+h)-f(a)}{h}.$$

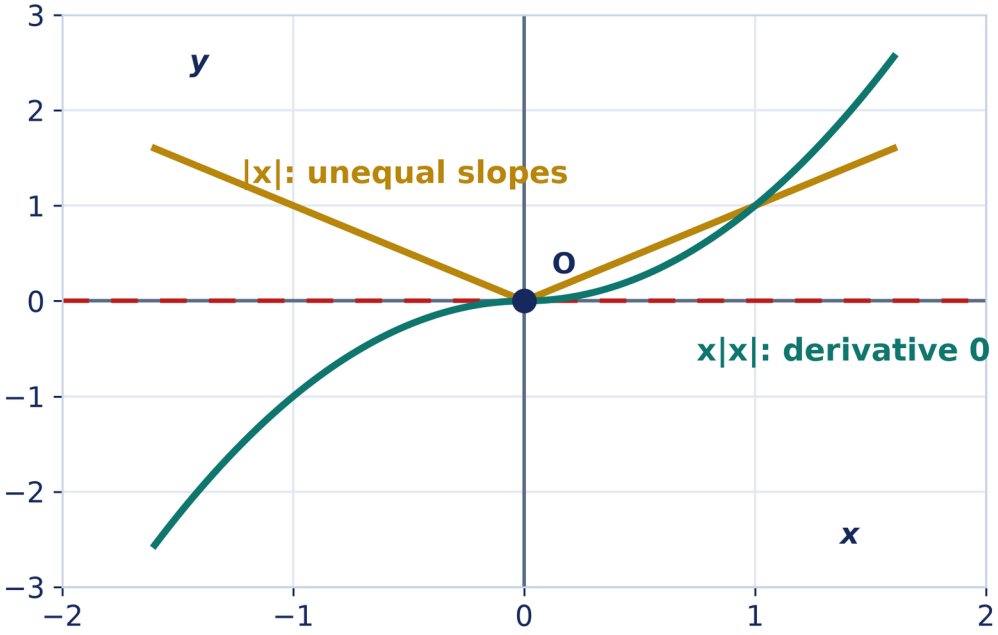
The two-sided derivative exists exactly when these are equal. On a domain with only one side, the corresponding one-sided derivative is the appropriate notion.

For $|x-2|$ at 2, the left derivative is -1 and the right derivative is 1, so the continuous corner is not differentiable. For $x|x|$ at 0,

$$\frac{h|h|}{h} = |h| \longrightarrow 0$$

from either side, so the derivative exists and equals 0.

A corner and a smooth flat join are different



READ THE FIGURE
Both functions are continuous at 0. The V-shaped graph has left/right slopes -1 and 1; the curved graph has difference quotient |h|, so both sides have slope 0. The red dashed line is its horizontal tangent.

Match values before matching derivative formulas

For

$$f(x) = x^2 \quad (x \le 1), \qquad f(x) = ax + b \quad (x > 1),$$

differentiability first requires continuity:

$$1 = a + b.$$

The left derivative at 1 is 2, while the right derivative is a. Therefore

$$a = 2, \qquad b = -1.$$

With these parameters the right formula is $2x - 1$, exactly the tangent to x^2 at 1.

WORKED EXAMPLE

A smooth junction needs two independent equations

Checking only $a = 2$ matches slopes but leaves $1 = 2 + b$, so continuity would fail unless $b = -1$. Checking only $a + b = 1$ matches values but does not control the slopes. Both conditions are necessary.

An infinite difference quotient is not a finite derivative

For $f(x) = x^{1/3}$ at 0,

$$\frac{f(h) - f(0)}{h} = h^{-2/3},$$

which is unbounded as $h \to 0$. The graph has a vertical tangent in a geometric sense, but $f'(0)$ does not exist as a finite real number.

COMMON TRAP

Do not differentiate each branch only away from the junction

Branch formulas tell you the one-sided derivatives only after the function value is known and continuous. Preserve the domain of each formula, then compare left limit, right limit, value, left derivative and right derivative explicitly.

LESSON 03 / 10

Linearity and the product rule come from the definition

Limit laws give $(af+bg)'=af'+bg'$. For a product, add and subtract $f(x+h)g(x)$:

$$\frac{f(x+h)g(x+h)-f(x)g(x)}{h}=\frac{f(x+h)-f(x)}{h}g(x)+f(x+h)\frac{g(x+h)-g(x)}{h}.$$

Taking $h\to 0$ proves

$$(fg)'=f'g+fg'.$$

The second term uses continuity of f, which follows from its differentiability.

WORKED EXAMPLE

Products with exact evaluations

For $x^2\sin x$,

$$(x^2\sin x)'=2x\sin x+x^2\cos x.$$

For $x^3\sin x$, the derivative is $3x^2\sin x+x^3\cos x$, so at $\pi/2$ it is $3\pi^2/4$. Also $(xe^x)'=e^x(1+x)$, whose value at 0 is 1.

Quotient and power rules need their domains

For $g(x)\neq 0$,

$$\left(\frac{f}{g}\right)'=\frac{f'g-fg'}{g^2}.$$

The power rule $(x^n)'=nx^{n-1}$ follows for positive integers by the binomial difference quotient or induction. Logarithmic differentiation later supplies the real-power version on the appropriate domain.

WORKED EXAMPLE

Two quotient calculations

$$\left(\frac{x}{1+x^2}\right)'=\frac{1-x^2}{(1+x^2)^2},\qquad \left(\frac{x^2}{x+1}\right)'=\frac{x^2+2x}{(x+1)^2}\quad (x\neq -1).$$

For $x>0$,

$$(x^5-3x^{1/2}+7)'=5x^4-\frac{3}{2\sqrt{x}}.$$

Choose the algebra rule and retain its hypothesis

Expression	Derivative rule	Essential condition
a f + b g	a f' + b g'	both derivatives exist
f g	f'g + fg'	both derivatives exist
f / g	(f'g - fg')/g^2	g(x) is nonzero
x^n	n x^(n-1)	use the valid power domain
1 / x	-1/x^2	x is nonzero

READ THE FIGURE

The quotient rule is not (f'/g'). Differentiate the numerator and denominator in the correct order and square the original denominator.

COMMON TRAP

Simplification can expose a domain restriction

Although

$$\frac{x^2}{x+1} = x - 1 + \frac{1}{x+1} \qquad (x \neq -1),$$

differentiating the simplified form still gives a derivative only on the original domain. Algebraic cancellation or division never restores the excluded point.

LESSON 04 / 10

Differentiate the outer function, then multiply by the inner rate

If g is differentiable at x and f is differentiable at $u = g(x)$, then

$$\frac{d}{dx} f(g(x)) = f'(g(x))g'(x).$$

A proof that also covers repeated zero inner increments writes $k = g(x+h) - g(x)$ and

$$f(u+k) - f(u) = (f'(u) + \varepsilon(k))k, \qquad \varepsilon(k) \to 0 \text{ as } k \to 0.$$

Since $k/h \to g'(x)$, division by h and passage to the limit give the chain rule without assuming $k \neq 0$ at every step.

Standard derivatives supply the outer rates

$$(\sin x)' = \cos x, \qquad (\cos x)' = -\sin x, \qquad (e^x)' = e^x, \qquad (\ln x)' = \tfrac{1}{x} \quad (x > 0).$$

For example, the sine formula follows by expanding $\sin(x+h)$ and using $\sin h/h \to 1$ and $(\cos h - 1)/h \to 0$. The logarithm formula follows from $\ln(1 + h/x) \sim h/x$.

WORKED EXAMPLE

Name the outer and inner functions explicitly

$$\frac{d}{dx}\sin(3x^2) = 6x\cos(3x^2), \qquad \frac{d}{dx}e^{x^2} = 2xe^{x^2},$$

and

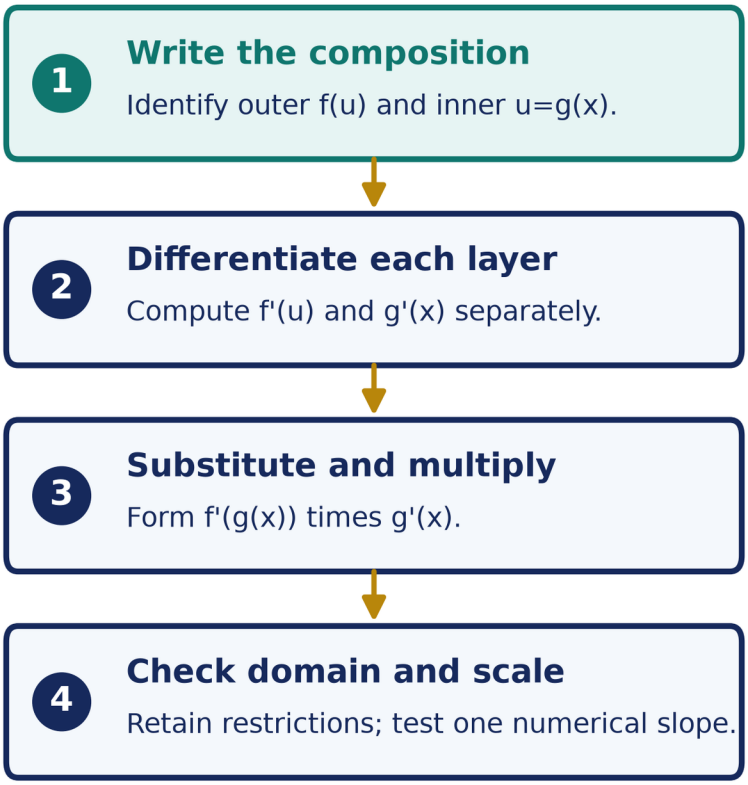
$$\frac{d}{dx}\cos(x^3) = -3x^2\sin(x^3).$$

Where $\sin x > 0$,

$$\frac{d}{dx}\ln(\sin x) = \frac{\cos x}{\sin x} = \cot x.$$

The derivative $\cot x$ is not claimed at zeros of $\sin x$, where the original logarithm is undefined.

A reliable chain-rule workflow



READ THE FIGURE

The outer derivative keeps the inner input unchanged. Only after evaluating that outer rate do you multiply by the inner derivative.

COMMON TRAP

Do not multiply by a missing inner derivative

The derivative of $\sin(3x)$ *is* $3\cos(3x)$*, not* $\cos(3x)$*. Conversely, the derivative of* $\sin x$ *has no extra factor. A useful check at* 0 *is that* $\sin(3h)/(3h) \to 1$*, so the slope of* $\sin(3x)$ *at* 0 *is* 3*.*

LESSON 05 / 10

The inverse-function rule is a reciprocal on the correct branch

If $y = f(x)$ has a local differentiable inverse and $f'(x) \neq 0$, differentiating $f(f^{-1}(y)) = y$ gives

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

The nonzero derivative ensures a nonvertical tangent for the inverse relation. Applying this to the principal trigonometric branches gives

$$(\arcsin x)' = \frac{1}{\sqrt{1-x^2}} \quad (|x| < 1),$$

$$(\arccos x)' = -\frac{1}{\sqrt{1-x^2}} \quad (|x| < 1), \qquad (\arctan x)' = \frac{1}{1+x^2}.$$

Thus $(\arctan x)'|_{x=1} = 1/2$, and for $|x| < 1/2$,

$$\frac{d}{dx}\arcsin(2x) = \frac{2}{\sqrt{1-4x^2}}.$$

Simplify first, but keep the principal branch visible

Where $1 + \cos x \neq 0$,

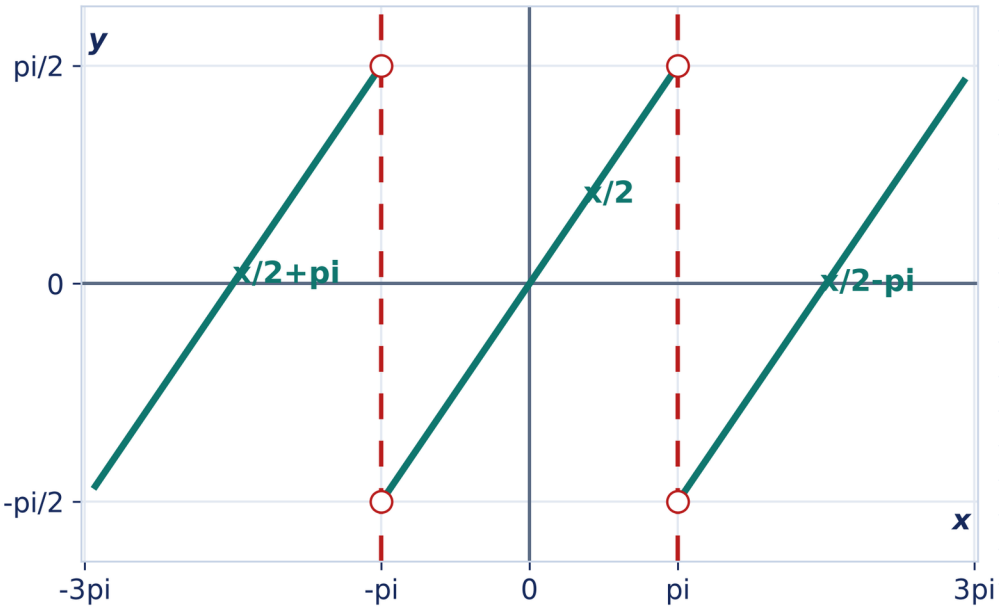
$$\frac{\sin x}{1+\cos x} = \tan\frac{x}{2}.$$

However, $\arctan(\tan u) = u$ only when $u \in (-\pi/2, \pi/2)$. Therefore

$$\arctan\left(\frac{\sin x}{1+\cos x}\right) = \frac{x}{2} \quad \text{only for } -\pi < x < \pi.$$

On each interval $((2k-1)\pi, (2k+1)\pi)$, the simplified value is $x/2 - k\pi$; the derivative is $1/2$ on every such interval.

arctan(tan(x/2)) repeats the principal branch



READ THE FIGURE

Each branch has slope 1/2, but the intercept changes by pi. At odd multiples of pi the original denominator 1+cos(x) is zero, so no value or derivative is claimed there.

COMMON TRAP

A trigonometric identity does not erase inverse-branch jumps

Replacing $\arctan(\tan(x/2))$ *by* $x/2$ *everywhere would invent values at odd multiples of* π *and remove the jumps between principal branches. State the interval before differentiating a simplified inverse-trigonometric expression.*

LESSON 06 / 10

Differentiate the relation, treating y as a function of x

If a local branch satisfies $F(x,y)=0$, differentiate both sides with respect to x. Every y-term uses the chain rule. When $F_y \neq 0$,

$$F_x + F_y\frac{dy}{dx} = 0 \quad\Longrightarrow\quad \frac{dy}{dx} = -\frac{F_x}{F_y}.$$

The point must first satisfy the original relation; the derivative formula alone does not prove that a branch exists.

WORKED EXAMPLE

A circle gives the slope without solving for y

For $x^2+y^2=25$,

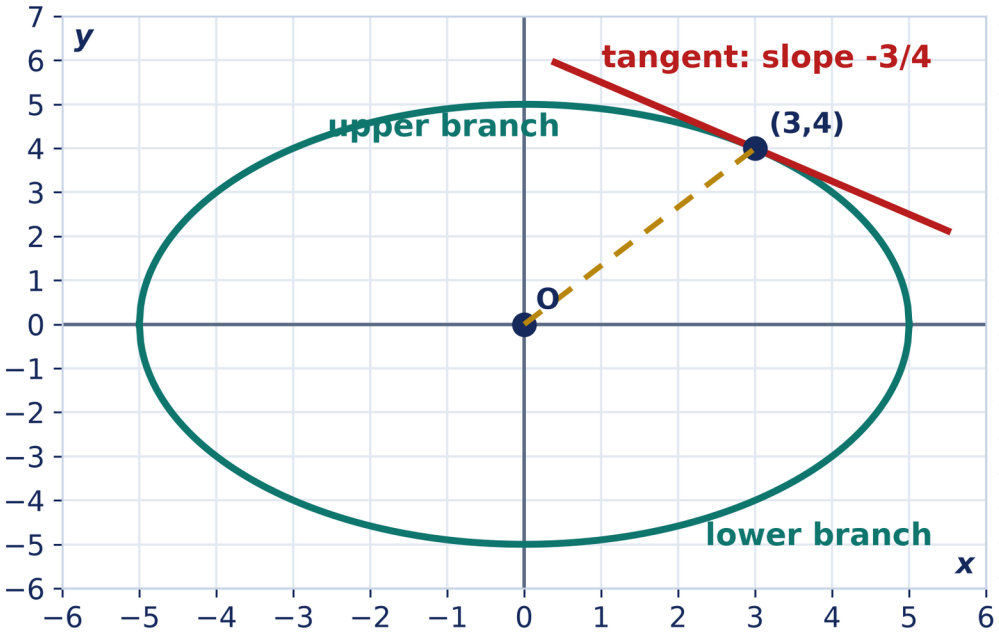
$$2x + 2y\frac{dy}{dx} = 0 \quad\Longrightarrow\quad \frac{dy}{dx} = -\frac{x}{y} \qquad (y\neq 0).$$

At $(3,4)$, the slope is $-3/4$. The tangent is

$$y - 4 = -\frac{3}{4}(x-3).$$

The radius has slope $4/3$, so the radius and tangent slopes multiply to -1.

The implicit tangent is perpendicular to the radius



READ THE FIGURE

The tangent uses only the local branch through (3,4). Its slope -3/4 is perpendicular to the radius slope 4/3.

WORKED EXAMPLE

Solve for the derivative only after collecting its coefficient

For $y^3 + xy = 2$ at $(1, 1)$,

$$3y^2y' + y + xy' = 0.$$

Thus $(3y^2 + x)y' = -y$, and substitution of the point gives

$$y' = -\tfrac{1}{4}.$$

The relation checks first: $1^3 + 1 \cdot 1 = 2$.

COMMON TRAP

The derivative formula can be undefined at a valid point

At $(5, 0)$ on the circle, $-x/y$ is undefined and the tangent is vertical. This does not contradict the relation; it means the local branch cannot be written as a differentiable function $y(x)$ there. Use x as a function of y or a parameter instead.

LESSON 07 / 10

Turn products, quotients and variable powers into sums

If $y > 0$ and y is differentiable, then $(\ln y)' = y'/y$. Thus logarithmic differentiation is the chain rule applied before multiplying back by y. It is ideal for $u(x)^{v(x)}$ and long products.

For a variable power with $u(x) > 0$,

$$y = u^v \quad \Longrightarrow \quad \ln y = v \ln u \quad \Longrightarrow \quad \frac{y'}{y} = v' \ln u + v\frac{u'}{u}.$$

WORKED EXAMPLE

Differentiate x^x without treating either x as constant

For $x > 0$ and $y = x^x$,

$$\ln y = x \ln x \quad \Longrightarrow \quad \frac{y'}{y} = \ln x + 1.$$

Therefore

$$y' = x^x(1 + \ln x), \qquad y'(1) = 1.$$

Neither the constant-base power rule nor the constant-exponent rule applies to x^x by itself.

WORKED EXAMPLE

Logarithms compress a product and a radical denominator

For an interval on which

$$y = \frac{x^2 \sin x}{\sqrt{1+x}}$$

is positive, logarithmic differentiation gives

$$\frac{y'}{y} = \frac{2}{x} + \cot x - \frac{1}{2(1+x)}.$$

Multiply by the original y at the end. On intervals where a nonzero product changes sign, use $\ln|y|$ and track each factor's domain instead of assuming positivity.

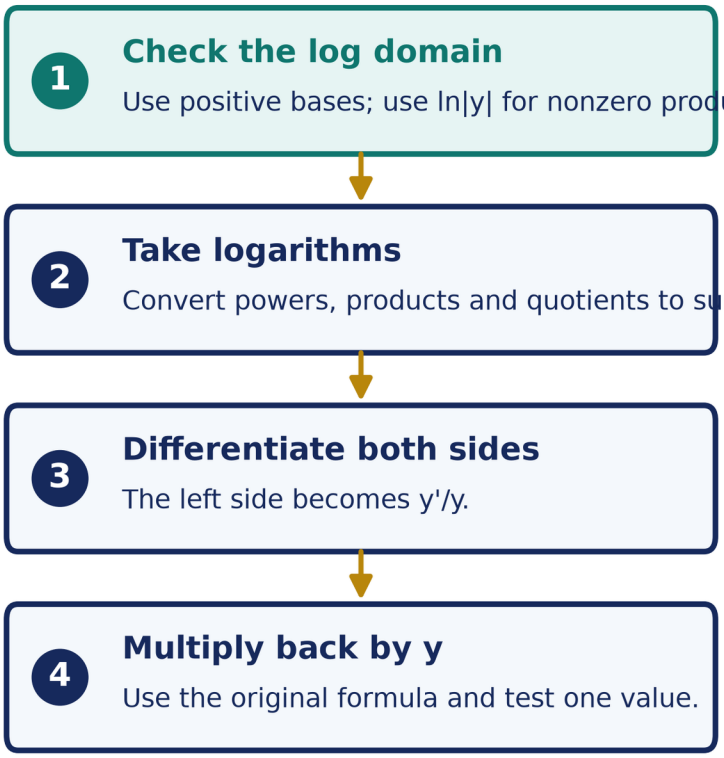
The same idea proves the real-power rule

For $x > 0$ and fixed real r, write $y = x^r = e^{r \ln x}$. Then

$$y' = e^{r \ln x}\frac{r}{x} = r x^{r-1}.$$

At other points, use the domain of the particular real power rather than extending this proof blindly.

Logarithmic differentiation preserves a four-step chain



READ THE FIGURE

Dropping the final factor y is the most common algebraic slip. The computed quantity before that step is the relative rate y'/y.

COMMON TRAP

Real variable powers need a positive base

The proof of $(u^v)' = u^v(v' \ln u + vu'/u)$ requires $u > 0$ near the point. A negative or zero base needs a separate definition and cannot be substituted into $\ln u$.

LESSON 08 / 10

Divide rates with respect to the same parameter

If $x = x(t)$, $y = y(t)$ and $dx/dt \neq 0$, then

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

For a second derivative, differentiate the **first derivative with respect to t**, then divide by dx/dt again:

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{dx/dt}.$$

It is generally wrong to compute $y''(t)/x''(t)$.

WORKED EXAMPLE

A cubic parameter gives both derivatives exactly

For $x = t^2$, $y = t^3$ and $t \neq 0$,

$$\frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3t}{2}.$$

Thus the slope at $t = 2$ is 3, and

$$\frac{d^2y}{dx^2} = \frac{3/2}{2t} = \frac{3}{4t},$$

whose value at $t = 1$ is $3/4$.

WORKED EXAMPLE

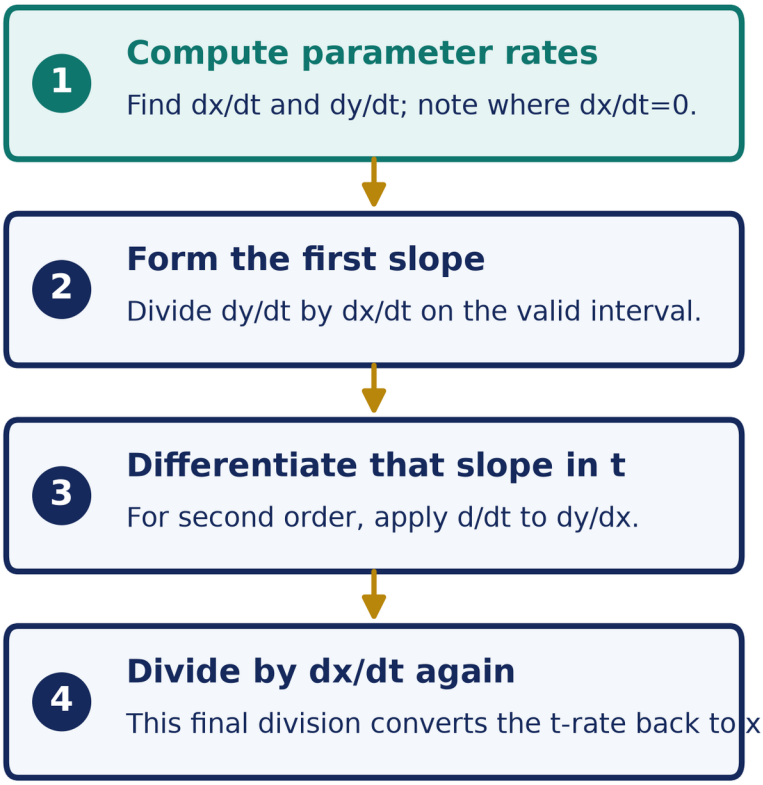
An ellipse parameter retains a trigonometric slope

For $x = a\cos t$, $y = b\sin t$ with $a, b \neq 0$ and $\sin t \neq 0$,

$$\frac{dy}{dx} = \frac{b\cos t}{-a\sin t} = -\frac{b}{a}\cot t.$$

At parameter values where $dx/dt = 0$, investigate a vertical tangent or another parametrization instead of cancelling $0/0$.

Parametric derivatives alternate between t and x



READ THE FIGURE

The derivative dy/dx is a function of the parameter. A second x-derivative therefore needs one more parameter derivative and one more division by dx/dt.

COMMON TRAP

A zero parameter rate needs a separate limiting analysis

At $t = 0$ for $x = t^2$, $y = t^3$, the displayed ratio is $0/0$. The simplified slope $3t/2$ still has limit 0, but this is a conclusion from taking a limit, not from substituting into the original quotient.

LESSON 09 / 10

Iterate derivatives and look for a cycle

Second and higher derivatives are defined by repeated differentiation:

$$f'' = (f')', \qquad f^{(n+1)} = (f^{(n)})'.$$

The trigonometric cycle

$$\sin x \to \cos x \to -\sin x \to -\cos x \to \sin x$$

has period 4. Hence $(\sin x)^{(4)} = \sin x$ and $(\sin x)''|_{x=\pi/2} = -1$. Also

$$\frac{d^n}{dx^n} e^{ax} = a^n e^{ax},$$

so the third derivative of e^{2x} at 0 is 8.

A normal uses the negative reciprocal of a nonzero tangent slope

At a point with $f'(a) = m \neq 0$, the normal slope is $-1/m$. For $y = x^3$ at $(1, 1)$, the tangent slope is 3, so the normal is

$$y - 1 = -\tfrac{1}{3}(x - 1).$$

Horizontal and vertical tangent cases are handled geometrically in Chapter 19 rather than by forcing the reciprocal formula.

Match the expression to the smallest sufficient method

Expression or question	Method	Final check
Definition at a point	Difference quotient	finite two-sided limit
Absolute or piecewise join	One-sided derivatives	value and slopes agree
Sum, product or quotient	Derivative algebra	denominator and power domain
Nested expression	Chain rule	multiply every inner rate
Inverse trigonometric form	Inverse rule or simplify	principal branch interval
Relation F(x,y)=0	Implicit differentiation	point lies on relation
Variable power or product	Logarithmic differentiation	positive base; multiply by y
x(t), y(t)	Parametric ratios	dx/dt nonzero; divide again for y''
nth derivative of product	Leibniz extension	keep only surviving orders

READ THE FIGURE

A correct rule applied to the wrong domain is still an incorrect answer. Check the expression, the point and the requested derivative order.

Verify a computed derivative with a finite difference

For small nonzero h,

$$\frac{f(x+h)-f(x)}{h}$$

should approach the claimed $f'(x)$. This is a numerical diagnostic, not a replacement for the exact simplification. For example, at $x=1$ the derivative of x^x is 1, and its forward difference quotients approach 1 from above.

KEY RESULT

Final differentiation self-check

Did I prove existence at a junction before using branch formulas? Did every inner derivative appear? Are inverse-trigonometric and logarithmic domains stated? For parametric second derivatives, did I divide by dx/dt twice in the correct places? For a normal, is the tangent slope nonzero? Can one exact value and one finite difference independently confirm the result?

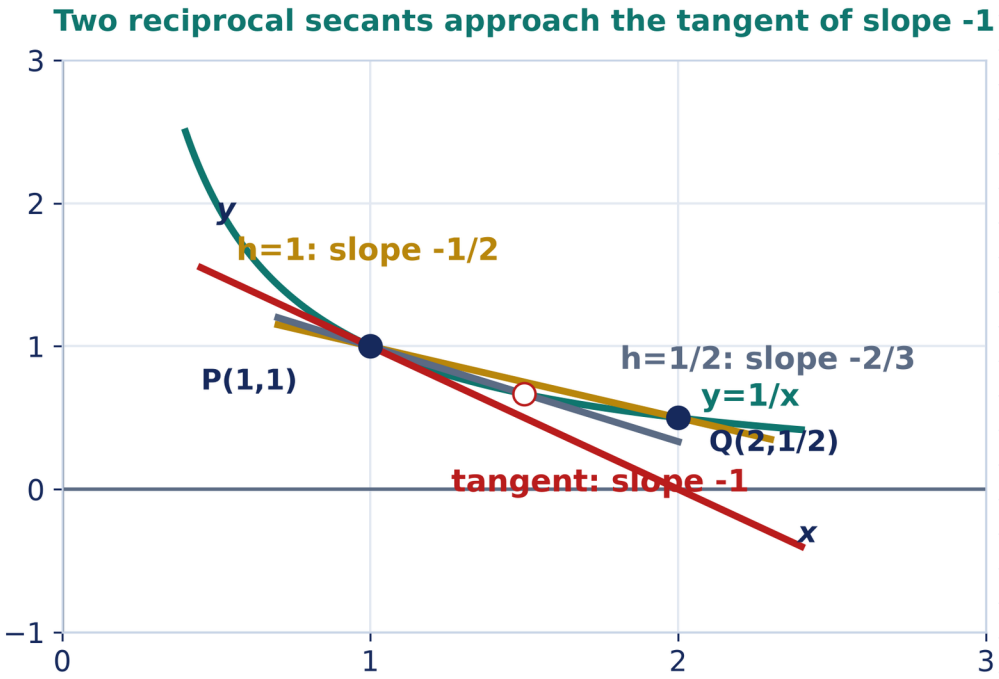
LESSON 10 / 10

Secant slopes reveal the derivative of a reciprocal

For $f(x) = 1/x$ at $P = (1, 1)$, the derivative is -1. The secants using $h = 1$ and $h = 1/2$ have slopes

$$\frac{1/2 - 1}{2 - 1} = -\frac{1}{2}, \qquad \frac{2/3 - 1}{1/2} = -\frac{2}{3}.$$

They approach the tangent slope -1 as $h \to 0$.



READ THE FIGURE

The secant slopes -1/2 and -2/3 are finite-difference approximations. The tangent y=2-x has the exact derivative slope -1 at P.

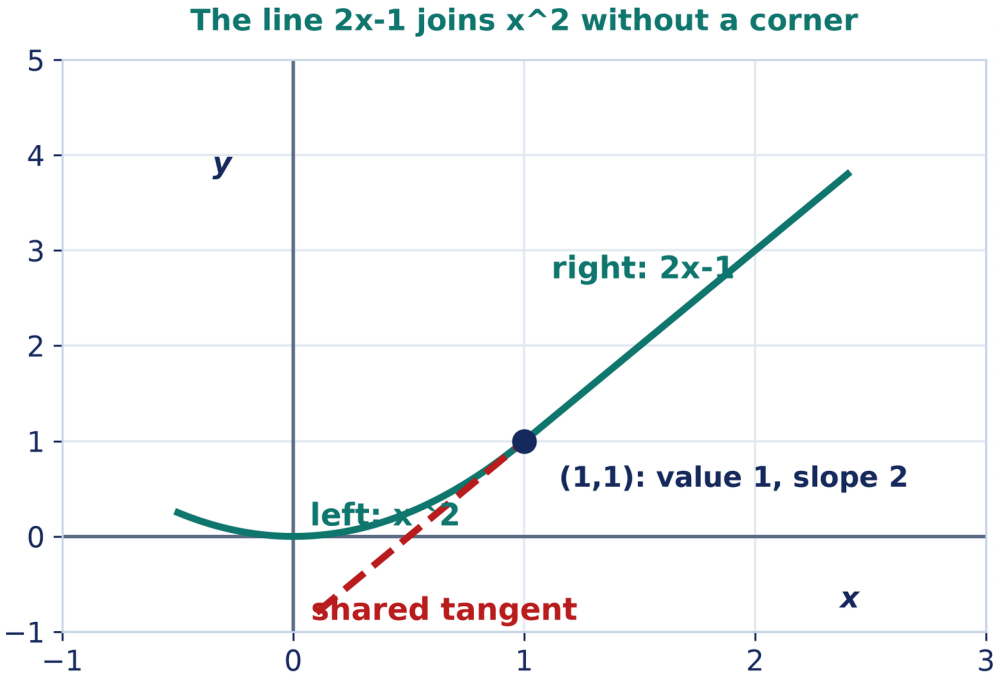
Visual check: the tangent and both secants pass through P. Their slopes are ordered $-1/2 > -2/3 > -1$, consistent with the convex reciprocal graph on $x > 0$.

A differentiable piecewise join matches value and slope

For x^2 on the left and $ax+b$ on the right of 1, the requirements are

$$a+b=1, \qquad a=2.$$

Thus $a=2$, $b=-1$, and the right branch is the tangent line $2x-1$.



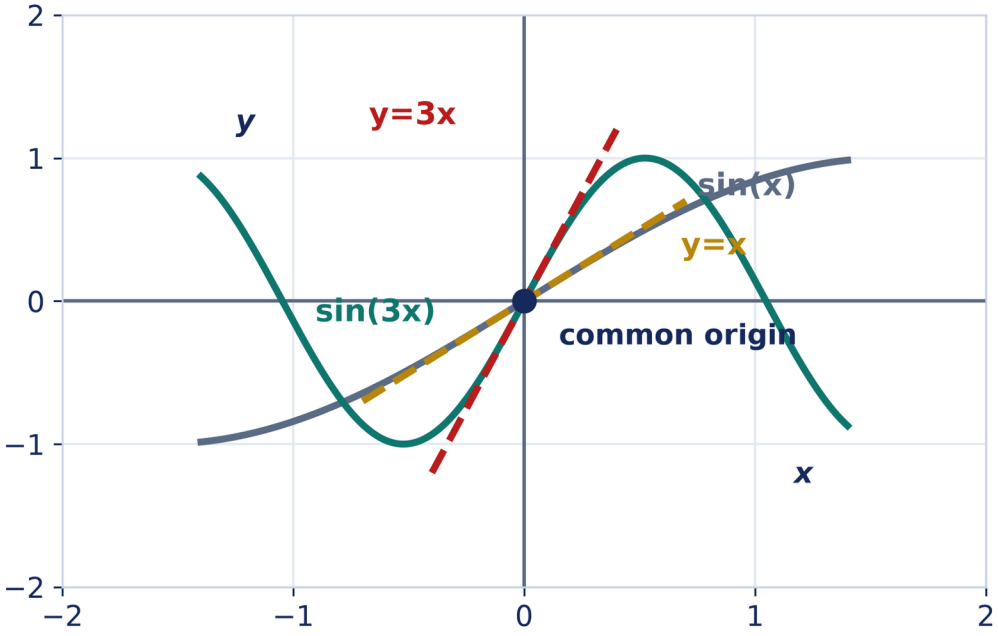
READ THE FIGURE
The function value is continuous and both one-sided derivatives equal 2. Merely drawing a connected graph would not be enough; the common tangent certifies the slope match.

Visual check: at 1, $x^2=1$ and $2x-1=1$. Their derivatives are $2x|_{x=1}=2$ and 2, so both differentiability conditions hold exactly.

Chain rule scales the slope at the origin

Near 0, $\sin x \approx x$ *while* $\sin(3x) \approx 3x$. *The same small-angle limit therefore gives tangent slopes* 1 *and* 3 *at the origin.*

The inner factor 3 triples the initial slope



READ THE FIGURE

Both sine curves pass through the origin, but sin(3x) oscillates three times as fast. Its local tangent is y=3x, illustrating the chain-rule factor 3.

Visual check: *for small* h, $\sin(3h)/h = 3\sin(3h)/(3h) \to 3$. *This agrees with* $(\sin 3x)' = 3\cos 3x$ *at* 0.

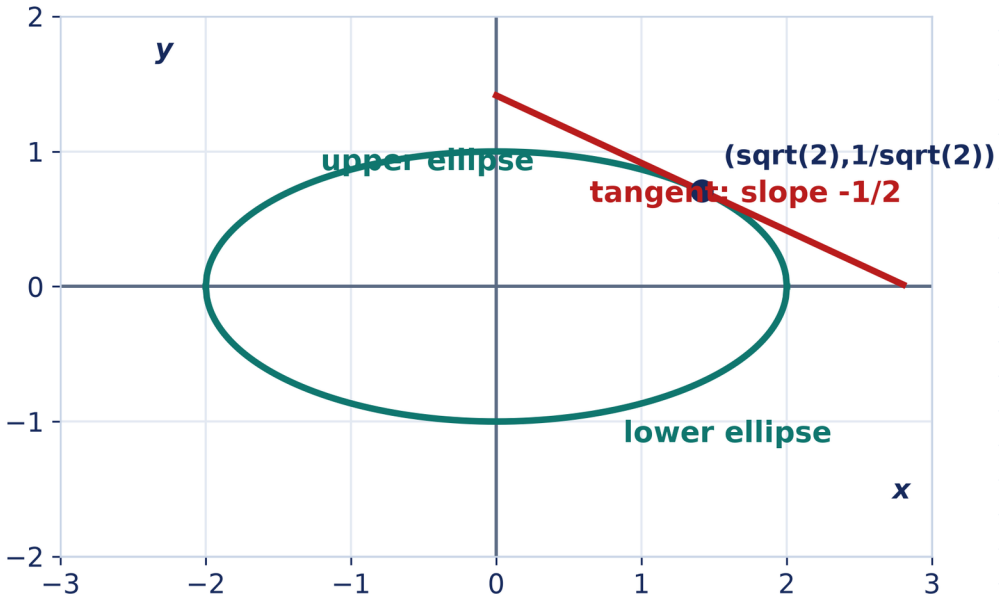
Implicit and parametric methods agree on an ellipse

For $x^2/4 + y^2 = 1$,

$$\frac{x}{2} + 2yy' = 0 \quad \Longrightarrow \quad y' = -\frac{x}{4y}.$$

With $x = 2\cos t$, $y = \sin t$, the parametric slope is $-(1/2)\cot t$. At $t = \pi/4$, the point is $(\sqrt{2}, 1/\sqrt{2})$ and both methods give slope $-1/2$.

Two differentiation methods produce the same ellipse tangent



READ THE FIGURE
The tangent is y=-x/2+sqrt(2). Implicit differentiation uses the relation; parametric differentiation uses t=pi/4. Their agreement is an independent check.

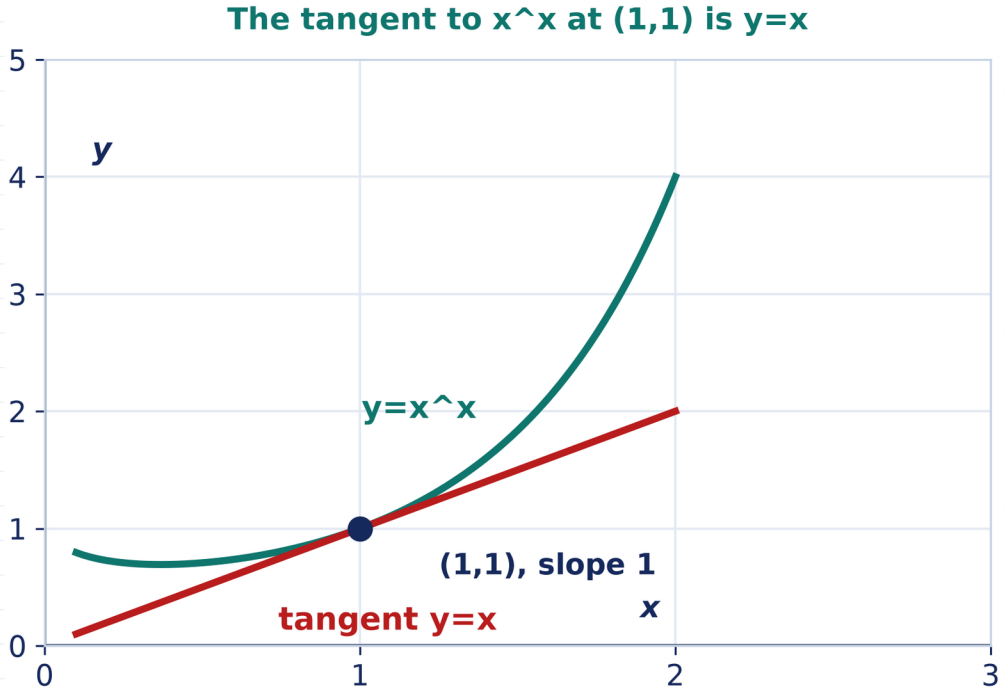
Visual check: substituting the tangent point gives $x^2/4 + y^2 = 1/2 + 1/2 = 1$. The tangent slope satisfies $-x/(4y) = -1/2 = -(1/2)\cot(\pi/4)$.

Logarithmic differentiation finds the tangent to x^x

For $x>0$, $y=x^x$ gives $\ln y = x\ln x$ and

$$y' = x^x(1+\ln x).$$

At $(1,1)$ the slope is 1, so the tangent is $y=x$.



READ THE FIGURE
Taking logarithms differentiates both occurrences of x at once. The original curve and the line y=x share the point and first derivative at 1.

Visual check: $1^1=1$ and $1^1(1+\ln 1)=1$. Forward difference quotients at 1 approach the same slope from above.

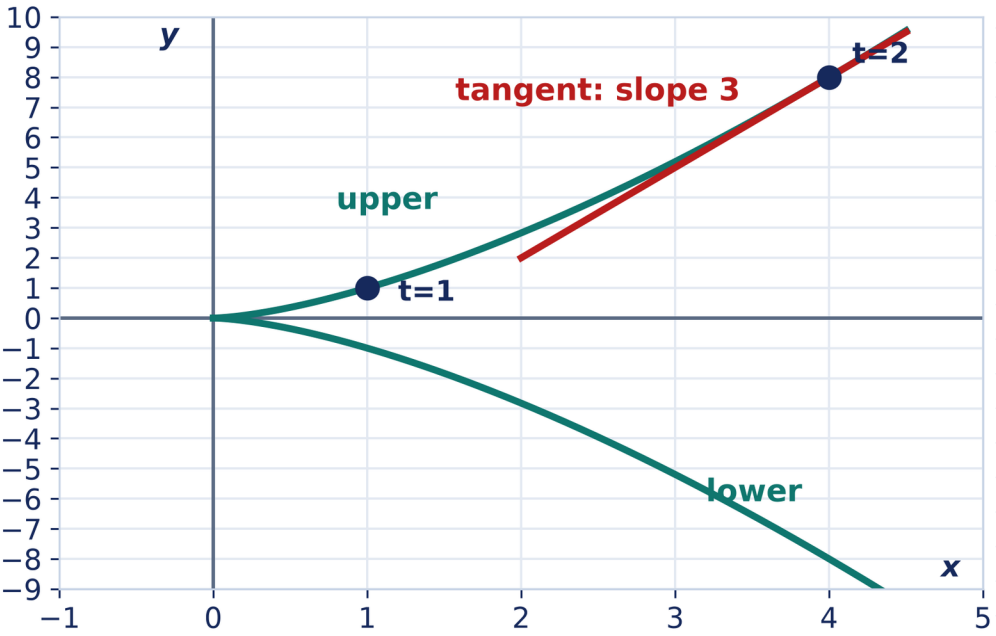
A second parametric derivative needs one more division

For $x = t^2$, $y = t^3$,

$$\frac{dy}{dx} = \frac{3t}{2}, \qquad \frac{d^2y}{dx^2} = \frac{3}{4t} \qquad (t \neq 0).$$

At $t = 1$, these are $3/2$ and $3/4$. At $t = 2$, the point is $(4, 8)$ and the tangent slope is 3.

The semicubical parabola has slope 3 at t=2



READ THE FIGURE

The curve is y^2=x^3 with two branches. The tangent at t=2 is y=3x-4. The second derivative 3/(4t) is not y''(t)/x''(t).

KEY RESULT

Read each figure as a derivative equation

A secant shows a finite difference; a tangent shows its limit. A smooth join requires matching values and slopes. Inner scaling changes local rates. Implicit and parametric methods must agree on the same point. Every logarithmic or parametric calculation should end by multiplying or dividing by the factor required by the chain rule.